

ANN PRACTICAL CHEAT SHEET (Simple Language)

1 Activation Functions

Purpose

Decides whether neuron should activate or not.

Types

Function	Range	Use
Sigmoid	0 to 1	Binary classification
ReLU	0 to ∞	Hidden layers
Tanh	-1 to 1	Better than sigmoid
Linear	Any value	Regression

Important

- Adds **non-linearity**
- Without activation $\rightarrow$ network behaves like linear model

Viva

 Why ReLU mostly used?

 Faster + avoids vanishing gradient

2 McCulloch Pitts Neuron (AND-NOT)

Simplest neuron model

Uses:

- Inputs
- Weights
- Threshold

Formula

Net = $\sum w_i x_i$

If $\text{net} \geq \text{threshold} \rightarrow 1$
Else $\rightarrow 0$

Important

- Binary inputs only
- Works like logic gates

Viva

 Why negative weight used?
 For NOT operation

Perceptron

 Single layer neural network

Used for:
 Binary classification

Formula

[
 $\text{Net} = wx + b$
]

Weight Update

[
 $w = w + n(\text{error})x$
]

Important

- Supervised learning
- Works only for linearly separable data

Viva

 Can perceptron solve XOR?
 No

Decision Boundary

Line separating classes

Equation:

$$[w_1x_1 + w_2x_2 + b = 0]$$

Important

- Straight line in perceptron
- Shows classification regions

Viva

 Why perceptron works here?
 Data is linearly separable

5 BAM (Bidirectional Associative Memory)

Stores pattern pairs

$$[X \leftrightarrow Y]$$

Important

- Bidirectional recall
- Uses bipolar values (-1,+1)

Weight Formula

$$[W = \sum X^T Y]$$

Viva

 Difference between BAM & Hopfield?
 BAM has 2 layers

6 Hopfield Network

👉 Associative memory network

Used to:

✅ Recover noisy patterns

🔑 Important

- Recurrent network
- Bipolar values
- Stable states

Weight Formula

$$W = \sum p p^T$$

Viva

? Why diagonal = 0?

✅ No self connection

7 XOR Problem

👉 XOR is NOT linearly separable

So:

- Perceptron ❌ fails
- MLP ✅ solves

🔑 Important

Needs:

✅ Hidden layer

Viva

? Why hidden layer needed?

✅ To learn nonlinear patterns

8 Backpropagation

👉 Learning algorithm for MLP

Steps:

1. Forward propagation
2. Calculate error
3. Backpropagation
4. Update weights

🔑 Important

Reduces error using gradients

Viva

? What is epoch?

✅ One full training cycle

9 CNN (Convolutional Neural Network)

👉 Used for image processing

Layers

Layer	Work
Conv2D	Feature extraction
Pooling	Reduce size
Flatten	Convert 2D $\rightarrow$ 1D
Dense	Classification

🔑 Important

Best for images

Viva

? Why CNN better than ANN for images?

✅ Preserves spatial information

10 MNIST Digit Recognition

👉 Handwritten digit classification

Dataset:

- 28×28 images
- Digits 0–9

🔑 Important

Uses:

- CNN
- Softmax output

Viva

? Why reshape image?
✅ CNN expects 3D input

1 1 Softmax Function

Formula

$$P_i = \frac{e^{x_i}}{\sum e^{x_j}}$$

👉 Converts output into probabilities

🔑 Important

Used in:

✅ Multi-class classification

1 2 Logistic Regression vs Neural Network

Logistic Regression Neural Network

Simple	Complex
Linear	Nonlinear
Faster	More accurate

Viva

- ? Why NN better?
 - ✓ Learns complex patterns
-

1 3 Important Terms

Term	Meaning
Epoch	One complete training cycle
Bias	Shifts decision boundary
Learning Rate	Speed of learning
Weight	Importance of input
Loss Function	Measures error
Gradient Descent	Minimizes loss

1 4 Most Important Viva Questions

- ? What is ANN?
 - ✓ Computational model inspired by human brain
-

? What is activation function?

- ✓ Decides neuron output
-

? What is backpropagation?

- ✓ Method to update weights using error
-

? Difference between ANN & CNN?

- ✓ CNN is specialized for images
-

? What is overfitting?

- ✓ Model memorizes training data
-

? What is dropout?

- ✓ Randomly removes neurons to avoid overfitting
-

LAST 1-MINUTE REVISION

- ✓ Perceptron → linear classifier
 - ✓ XOR → needs hidden layer
 - ✓ Backpropagation → weight update
 - ✓ CNN → image processing
 - ✓ ReLU → most used activation
 - ✓ Softmax → probabilities
 - ✓ Hopfield/BAM → associative memory
 - ✓ Sigmoid → binary output
-



GOLDEN FORMULAS

GOLDEN FORMULAS

Perceptron

$$Net = wx + b$$

Sigmoid

$$\frac{1}{1 + e^{-x}}$$

ReLU

$$\max(0, x)$$

Softmax

$$\frac{e^x}{\sum e^x}$$

Weight Update

$$w = w + \eta(\text{error})x$$

FINAL EXAM TIP

If examiner asks:

👉 “Explain flow”

Say:

✅ Input → Forward Propagation → Error Calculation → Backpropagation → Weight Update → Output

That single line works for MOST ANN programs 🚀